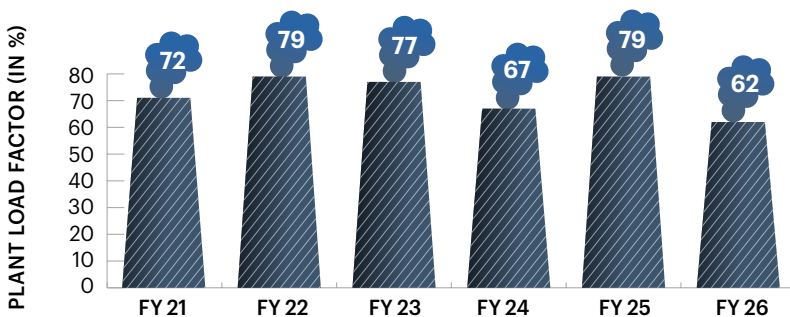


COUNTING THE ASSETS

- India's nuclear generation has remained largely stable
- Plant load factor has remained in the 65–85% range, indicating consistently high utilisation

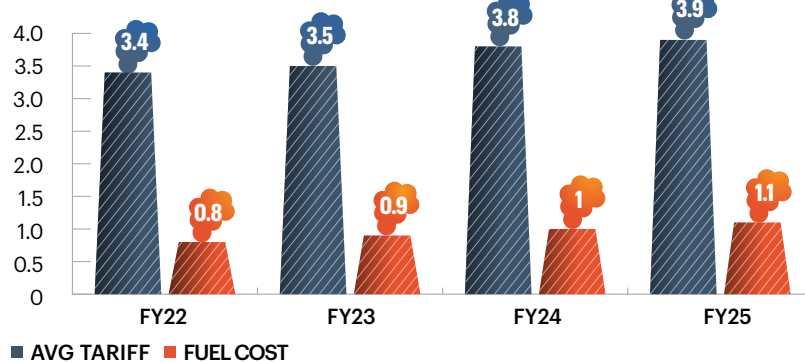


NOTE PLANT LOAD FACTOR IS THE RATIO OF THE AVERAGE POWER GENERATED TO ITS MAXIMUM INSTALLED CAPACITY **SOURCE** CEA

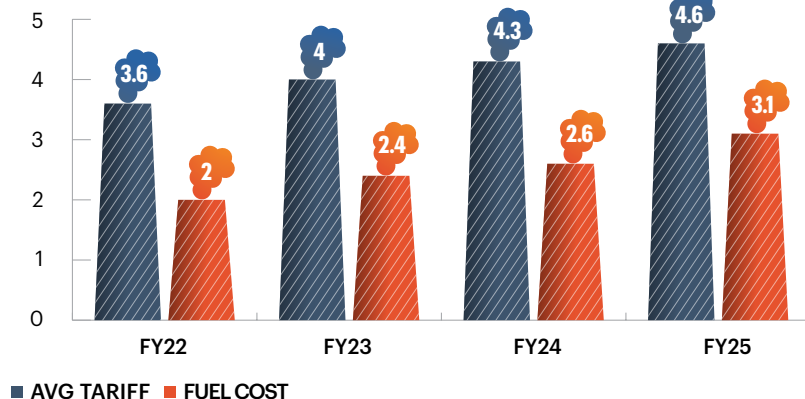
LOW COST

- A comparison of the average tariff and fuel cost of NPCIL and NTPC shows that nuclear power is more financially feasible

NPCIL (₹/UNIT)



NTPC (₹/UNIT)



SOURCE PRABHUDAS LILLADHAR

pound interest problem, and India is impatient,” says Prabhat Ranjan, Vice Chancellor of D Y Patil International University, who has been doing research in nuclear power. He is also the co-founder of ASPL Fusion, a private Indian deep-tech start-up focused on nuclear fusion.

ROLE OF START-UPS

India has been working on technology for thorium-based reactors for decades. Now, a US-based private start-up, Clean Core Thorium Energy (CCTE), has created a new type of fuel that blends thorium with a more concentrated type of uranium called HALEU (high-assay low-enriched uranium).

This blended fuel can be used in India's PHWRs, which make up the bulk of the country's existing nuclear power capacity and many of the new units under development now. National Thermal Power Corporation (NTPC) has tied up with CCTE for indigenisation of fuel manufacturing.

Kakodkar says that CCTE's technology can be used as an interim measure, where we generate power from thorium-uranium and generate fuel for stage 3, where we use thorium.

“Introduce the thorium-uranium-based fuel in stage one of the new capacity of the operational PHWRs. By irradiating thorium, you start building Uranium 233, not found naturally. It comes out as spent fuel inventory and is needed as fuel for Stage 3 of the nuclear programme. We can use this route also for energy independence as a bridge. So, this is a good method to lift to stage three, while stage two development can continue,” he says.